

Roll No:

Total Printed Pages: 2

AM100101

CT- II (December 25)

Subject- Applied Maths-I

B. TECH – I SEM (Q1 - GROUP)

Maximum Marks: 40

Min Marks- 14

Duration: 2 hours

Note:-

- 1) All questions are compulsory.

Section -1

Q. (1) Objective Questions:

(1x4=4)

- (a) Find the Jacobian of the following transformation:

$$u = ax+by, v = cx+dy$$

- (b) If $r = f(\theta)$ then length =

(c) $\int_0^{\frac{\pi}{2}} \cos^6 \theta = \dots \dots \dots$

- (d) Find the Complementary Function of:

$$(D^2+2D+1)y = 0.$$

Q. (2) Short Answer Type Questions:

(4x3=12)

1. Find the maximum and minimum value of:

$$u = x^3 - 4xy + 2y^2$$

2. Find the reduction formula of: $\int \tan^n x dx$.

3. Find the whole area of the cardioid, $r = a(1+\cos \theta)$.



Section -2

Q. (3) Descriptive Answer Type Questions:

(8×3=24)

1. The temperature T at any point (x, y, z) in space is $T = 400xyz^2$. Find the highest temperature on the surface of the unite sphere $x^2 + y^2 + z^2 = 1$.

Or

Find the volume of a sphere of a radius a .

2. Solve: $(D^2 - 2D + 1)y = xe^x \sin x$.

Or

Prove that:

$$\int_0^{\pi} \cos^n x \cos nx dx = \frac{\pi}{2^{n+1}}$$

$P.T.:$ $\frac{\pi}{2^{n+1}}$

3. Apply the method of variation of parameters to solve:

$$\frac{d^2 y}{dx^2} + 4y = \tan 2x.$$

Or

Solve. $(2x + 3)^2 \frac{d^2 y}{dx^2} - (2x + 3) \frac{dy}{dx} - 12y = 6x$

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CT- I (October 25)

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B. TECH – I SEM (Q1 - GROUP)

Maximum Marks: 40

Min Marks- 14

Duration: 2 hour

Note:-

- 1) All questions are compulsory.

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Section -1

Q. (1) Objective Questions:

(1x4=4)

- (a). Define trace of a matrix with example.
(b). State Cayley Hamilton theorem.
(c). If $y = x^{\pi}$, find y_{xx} .
(d). State Euler's theorem on homogeneous function.

Q. (2) Short Answer Type Questions:

(4x3=12)

1. Find the rank of the matrix: $A = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 6 & 8 \\ 3 & 7 & 22 \end{bmatrix}$.

2. Apply MacLaurin's theorem to prove that:

$$\log(1 + e^x) = \log_e 2 + \frac{1}{2}x + \frac{1}{8}x^2 - \frac{1}{192}x^4 + \dots$$

3. If $u = e^{xyz}$, prove that: $\frac{\partial^3 u}{\partial x \partial y \partial z} = e^{xyz}(1 + 3xyz + x^2y^2z^2)$.



Section -2

Q. (3) Descriptive Answer Type Questions:

(8x3=24)

1. Find the characteristic equation of the matrix:

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \text{ and hence find its inverse } A^{-1}.$$

Or

Test for consistency and solve:

$$5x + 3y + 7z = 4$$

$$3x + 26y + 2z = 9$$

$$7x + 2y + 10z = 5.$$

2. If $y = \frac{1}{x^2+a^2}$ find y_n .

Or

If $y = (\sin^{-1}x)^2$, prove that:

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0.$$

3. Verify Euler's theorem for; $u = x^3 \log\left(\frac{y}{x}\right)$.

Or

If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that: $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$.

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